

1. Student Active Learning & Assessment Events

Coding Problem Submission
(Execution verdict, runtime, test passes)

AI Mock Interview Session
(Acoustic fluency, speech rate, demeanor index)

LMS Telemetry Stream
(Video pause/drop-off, reading time, login streak)

Delivers New Adaptive Tasks

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Adaptive AQG Assessment Item
(Selected option, response latency, hints used)

2. M10: Behavioral Telemetry & Analytics Engine

Event Ingestion Buffer
(Redis Stream / TimescaleDB Hypertable)

Behavioral Aggregator & Rollup Worker
- Rolling 7-Day Study Hours Calculation
- Login Frequency & Retest Persistence Metric
- Hesitation & Drop-off Index Formulation

Weeks 3-4 Early Warning Trigger
(Detects Disengagement Drop: Activity Decrease > 30%)

3. M01: Invariant 22-Dimensional SPV Update

Profile Vector Synchronization Worker
Atomically re-evaluates:
- F04 (LeetCode count) & F06 (DSA Exam score)
- F14-F17 (Mock score, WPM, fillers, confidence)
- F18-F22 (Study hours, logins, persistence, roadmap %)

PostgreSQL SPV Ledger
(Versioned Snapshot with Timestamp)

4. M11: Digital Twin Multi-Stakeholder Sync

Digital Twin Synchronization Engine
- Computes Delta: $\Delta v_s = v_s(t) - v_s(t-1)$
- Recalculates Historical Trajectory & Velocity
- Projects Future Placement Probability Horizon

Digital Twin State Store
(Real-Time Redis Cache + Historical PG)

5. Closed-Loop Adaptation (M02, M07, M08)

M02: Skill Gap Vector Update
(Removes mastered concepts, shrinks distance)

M07: A* DAG Dynamic Re-routing
(Re-computes optimal path if student struggles or excels)

M08: Adaptive Roadmap Revision
- Inserts prerequisite remedial nodes if failure rate > 40%
- Accelerates sprint schedule if mastery threshold met

6. Stakeholder Interventions & UI Updates

Student Dashboard UI
(Roadmap dynamically updates, next task ready)

Faculty / Mentor Portal
(At-Risk early warning alert triggered)

Recruiter Talent Pool
(Differential privacy aggregate readiness shifts)